



Organisation

Name: \_\_\_\_\_

Class: \_\_\_\_\_

Date: \_\_\_\_\_

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Time: **65 minutes**

Marks: **61 marks**

Comments:

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1.

Lipases break down lipids.

(a) Which **two** products are formed when lipids are broken down?

Tick (✓) **two** boxes.

Amino acids

☐

Fatty acids

☐

Glucose

☐

Glycerol

☐

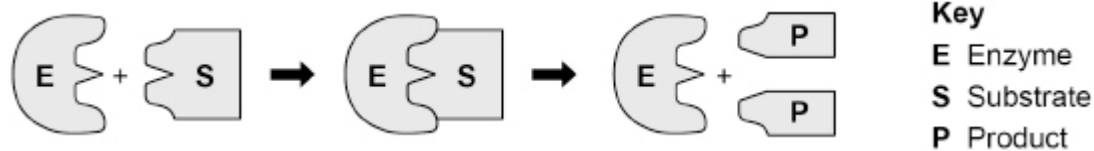
Glycogen

☐

(2)

One model used to explain enzyme action is the 'lock and key theory'.

The diagram below shows a model of the theory.



(b) Explain the 'lock and key theory' of enzyme action.

Use information from the diagram above in your answer.

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(3)

- (c) There are many different types of lipase in the human body.

Why does each different type of lipase act on only **one** specific type of lipid molecule?

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Students investigated the presence of starch and glucose in the leaves of geranium plants.

This is the method used.

- 1 Place two identical geranium plants on a bench near a sunny window for two days.
- 2 After two days:
  - leave one plant near the window for two more days.
  - place one plant in a cupboard with no light for two more days.
- 3 Remove one leaf from each plant.
- 4 Crush each leaf to extract the liquid from the cells.
- 5 Test the liquid from each leaf for glucose and for starch.

(1)

- (d) Describe how the students would find out if the liquid from the leaf contained glucose.

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(3)

- (e) Describe how the students would find out if the liquid from the leaf contained starch.

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(2)

The table below shows the students' results.

| Test    | Leaf from plant kept in light for four days | Leaf from plant kept in light for two days and then no light for two days |
|---------|---------------------------------------------|---------------------------------------------------------------------------|
| Glucose | Strong positive                             | Weak positive                                                             |
| Starch  | Positive                                    | Negative                                                                  |

- (f) Explain why the leaf in the light for four days contained both glucose and starch.

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(2)

- (g) Explain why the leaf left in a cupboard with no light for two days did contain glucose but did **not** contain starch.

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(3)

- (h) Suggest **one** way the students could develop the investigation to find out more about glucose and starch production in plants.

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(1)

(Total 17 marks)

2.

**Table 1** shows information about some food components in cow's milk.

**Table 1**

|                    | Value per<br>500 cm <sup>3</sup> | Recommended Daily<br>Allowance (RDA) for a<br>typical adult |
|--------------------|----------------------------------|-------------------------------------------------------------|
| Energy in kJ       | 1046                             | 8700                                                        |
| Fat in g           | 8.4                              | 70.0                                                        |
| Salt in g          | 0.5                              | 6.0                                                         |
| Calcium in mg      | 605                              | 1000                                                        |
| Vitamin B-12 in µg | 4.5                              | 2.4                                                         |

- (a) How much **more** milk would a typical adult have to drink to get their RDA for calcium compared with the amount of milk needed to get their RDA for vitamin B-12?

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Volume of milk = \_\_\_\_\_ cm<sup>3</sup>

(3)

- (b) Describe how a student could test cow's milk to show whether it contains protein and different types of carbohydrate.

This image shows a blank sheet of white paper with horizontal ruling lines. The lines are evenly spaced and extend across the width of the page. There are no margins, text, or other markings on the paper.

**(6)**

A scientist investigated the effect of bile on the breakdown of fat in a sample of milk.

The scientist used an indicator that is colourless in solutions with a pH lower than 10, and pink in solutions with a pH above 10.

This is the method used.

1. Add 1 drop of bile to a test tube and one drop of water to a second test tube.
2. Add the following to each test tube:
  - 5 cm<sup>3</sup> of milk
  - 7 cm<sup>3</sup> of sodium carbonate solution (to make the solution above pH 10)
  - 5 drops of the indicator
  - 1 cm<sup>3</sup> of lipase.
3. Time how long it takes for the indicator in the solutions to become colourless.

The results are shown in **Table 2**.

**Table 2**

|                              | Time taken for the indicator to become colourless in seconds |
|------------------------------|--------------------------------------------------------------|
| <b>Solution with bile</b>    | 65                                                           |
| <b>Solution without bile</b> | 143                                                          |

- (c) Explain why the indicator in both tubes became colourless.

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**(3)**

- (d) Give the reason why the measurement of the time taken for the indicator to become colourless might be inaccurate.

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**(1)**

- (e) Explain the difference in the results for the two test tubes in **Table 2**.

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(3)

(Total 16 marks)

3.

The circulatory system is composed of the blood, blood vessels and the heart.

- (a) Urea is transported in the blood plasma.

Name **two** other substances transported in the blood plasma.

1. \_\_\_\_\_
2. \_\_\_\_\_

(2)

- (b) Some athletes train at high altitude.

Training at high altitude increases the number of red blood cells per  $\text{cm}^3$  of blood.

Explain why having more red blood cells per  $\text{cm}^3$  of blood is an advantage to an athlete.

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(3)



(c) Which **two** blood vessels carry deoxygenated blood?

Tick **two** boxes.

Aorta

☐

Coronary artery

☐

Pulmonary artery

☐

Pulmonary vein

☐

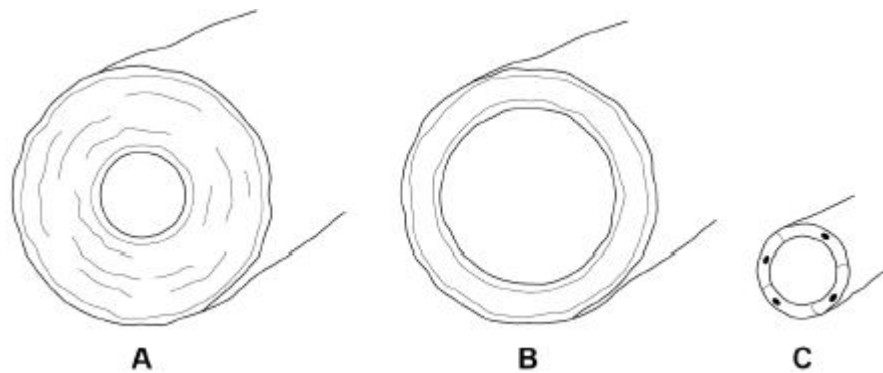
Vena cava

☐

(2)

**Figure 1** shows the three types of blood vessel.

**Figure 1**



(d) Which type of blood vessel carries blood into the right atrium?

Tick **one** box.

|   |  |
|---|--|
| A |  |
|---|--|

|   |  |
|---|--|
| B |  |
|---|--|

|   |  |
|---|--|
| C |  |
|---|--|

(1)

- (e) Compare the structure of an artery with the structure of a vein.

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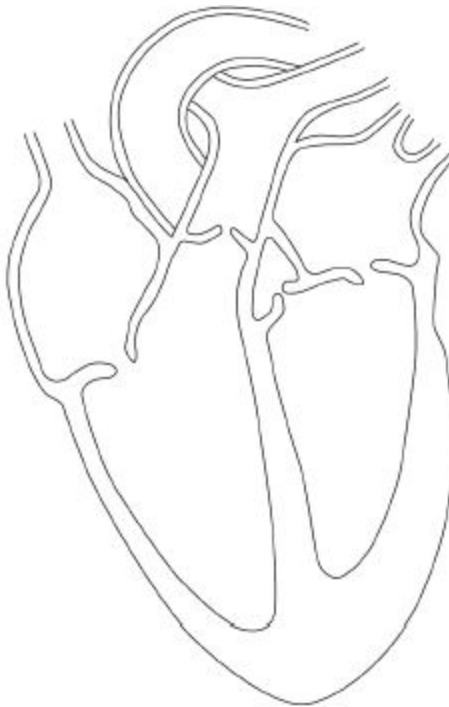
(3)

- (f) Heart rate is controlled by a group of cells. This group of cells act as a pacemaker.

**Figure 2** shows a section through the heart.

Draw an **X** on **Figure 2** to show the position of the pacemaker.

**Figure 2**



(1)

- (g) A patient may be fitted with an artificial pacemaker.

What condition may be treated using an artificial pacemaker?

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(1)

(Total 13 marks)

4.

Water moves from a plant to the atmosphere through the leaves.

- (a) How is the volume of water lost from the leaves controlled?

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(1)

- (b) Describe the transport of water through a plant from the roots to the atmosphere.

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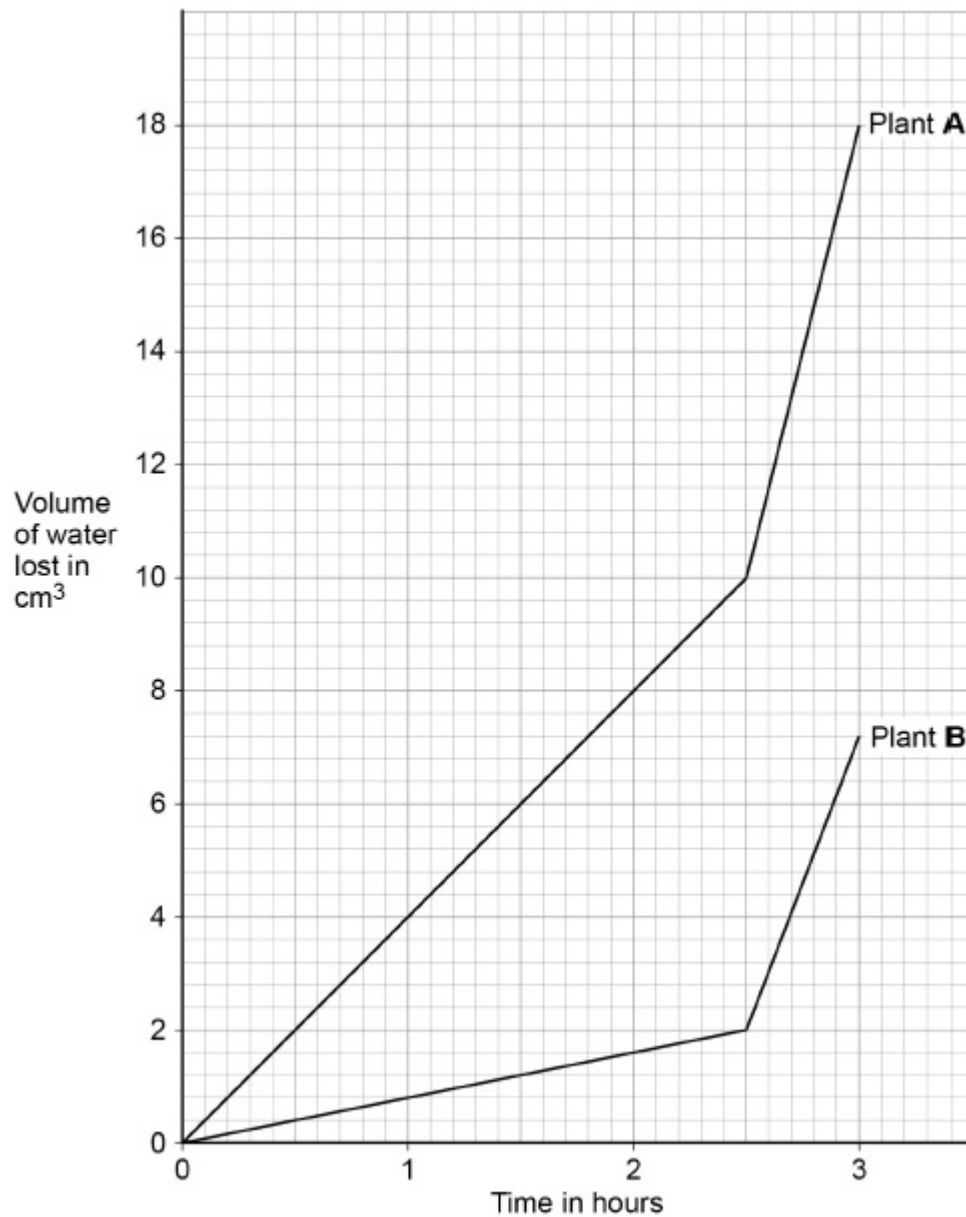
(3)

A student investigated the volume of water lost from two plants of different species.

Both plants were kept together.

**Figure 1** shows the student's results.

**Figure 1**



- (c) Suggest **one** reason for the difference in the rate of water loss from the two plants in the first 2.5 hours.

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(1)

Both plants were moved to a different place at 2.5 hours.

- (d) Calculate the rate of water loss per hour in plant **B** from 2.5 hours to 3 hours.

Give your answer to **2** significant figures.

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Rate of water loss = \_\_\_\_\_  $\text{cm}^3/\text{hour}$

(3)

- (e) Suggest **two** reasons why the rate of water loss in both plants changed after 2.5 hours.

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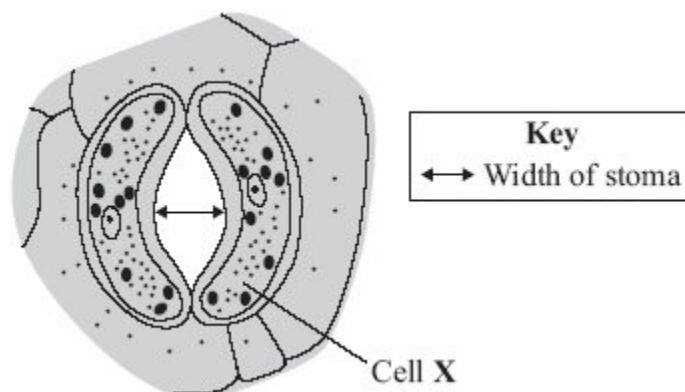
(2)

(Total 10 marks)

5.

Plant leaves have many stomata.

The diagram shows a stoma.



- (a) Name cell **X** \_\_\_\_\_

(1)

- (b) The table shows the mean widths of the stomata at different times of the day for two different species of plant.

Species **A** normally grows in hot, dry deserts.

Species **B** grows in the UK.

|       | Time of day<br>in hours | Mean width of stomata as a percentage of<br>their maximum width |           |
|-------|-------------------------|-----------------------------------------------------------------|-----------|
|       |                         | Species A                                                       | Species B |
|       | 0                       | 95                                                              | 5         |
| Dark  | 2                       | 86                                                              | 5         |
|       | 4                       | 52                                                              | 6         |
| Light | 6                       | 6                                                               | 40        |
|       | 8                       | 4                                                               | 92        |
|       | 10                      | 2                                                               | 98        |
|       | 12                      | 1                                                               | 100       |
|       | 14                      | 0                                                               | 100       |
|       | 16                      | 1                                                               | 96        |
|       | 18                      | 5                                                               | 54        |
| Dark  | 20                      | 86                                                              | 6         |
|       | 22                      | 93                                                              | 5         |
|       | 24                      | 95                                                              | 5         |

The data in the table show that species **A** is better adapted than species **B** to living in hot, dry deserts.

Explain how.

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**(4)**

**(Total 5 marks)**

## Mark schemes

1.

- (a) fatty acids

1

glycerol

1

- (b) enzyme binds to the substrate because they are complementary (shapes)

*allow enzyme joins to the substrate because they fit together exactly*

*allow enzyme joins to the substrate because the substrate fits the active site*

*ignore reference to specificity do **not** accept same shape*

1

(so) substrate is broken down (into products)

*allow (so) substrate splits (into products)*

*ignore products are formed, unqualified*

1

(so) products are released **or** enzyme is not changed

*allow enzyme is not used up*

*allow reference to activation energy for either marking point 2 **or** marking point 3*

1

- (c) each active site has a specific shape (so only fits one type of lipid molecule)

*allow each active site is a different shape*

*do **not** accept reference to the substrate having an active site*

1

- (d) add Benedict's (solution / reagent to the liquid)

1

boil / heat

*allow any temperature of 65 °C or above*

1

(if glucose is present the blue) colour changes to yellow / green / orange / brown / (brick) red

1

- (e) add iodine solution / reagent (to the liquid)

*allow add a drop of iodine*

*ignore iodine unqualified*

1

(if starch is present) it changes colour to blue / black (from yellow / orange / brown)

1



- (f) glucose from photosynthesis  
*do **not** accept starch made in photosynthesis*  
 1
- (excess) glucose converted to starch  
*allow (excess) glucose is stored as starch*  
 1
- (g) starch (stores) have been converted to glucose  
*ignore reference to residual glucose from previous photosynthesis*  
 1
- (so the glucose can be) used for respiration / (named) metabolic reactions  
**or** (so the glucose can be) used to release energy  
*do **not** accept idea of energy being produced / created / made*  
 1
- (because) there is no light to make (new / more) glucose by photosynthesis  
 1
- (h) any **one** from:
- test roots / stems of plants (in the light and dark)  
*do **not** accept reference to changing the independent variable*  
*allow test other parts of the plants*
  - test other species of plant  
*allow test other types of plant*
  - measure the concentrations of glucose **and** starch  
*ignore mass / amount*
  - vary the time in the dark / light
  - test variegated leaves  
*allow any other valid extension ignore repeats*
- 1

[17]

2.

(a) (for calcium)

$$\frac{500}{605} \times 1000 = 826.446281 (\text{cm}^3)$$

*allow any correct rounding to minimum 3 significant figures*  
*allow alternative route with correct rounding*

1

(for vitamin B-12)

$$\frac{500}{4.5} \times 2.4 = 266.67 (\text{cm}^3)$$

*allow alternative route with correct rounding*

1

560 / 559.8 / 559.78 / 559 ( $\text{cm}^3$ )

*allow only correct answer based on values given for vitamin B-12 and calcium*

1

*an answer of 560 / 559.8 / 559.78 / 559 ( $\text{cm}^3$ ) scores 3 marks*

*an incorrect answer for one step does not prevent allocation of marks for subsequent steps*

(b) **Level 2:** Scientifically relevant facts, events or processes are identified and given in detail to form an accurate account.

4–6

**Level 1:** Facts, events or processes are identified and simply stated but their relevance is not clear.

1–3

**No relevant content**

0

**Indicative content**

- Biuret reagent (allow  $\text{CuSO}_4$  and  $\text{NaOH}$ ) tests for protein
- add Biuret reagent to milk
- solution will turn (from blue) to lilac if positive
- iodine solution tests for starch (ignore iodine unqualified)
- add iodine solution to milk
- solution will turn (from orange / brown) to blue / black if positive
- Benedict's reagent tests for sugars
- add Benedict's reagent to milk and boil / heat (allow any temperature above  $60^\circ\text{C}$ )
- solution will turn (from blue) to (brick) red / brown / orange / yellow / green if positive

for **level 2**, reference to all three food tests is required

- (c) lipase breaks down fat into fatty acids (and glycerol)  
*do **not** accept if 'glycerol' is contradicted* 1
- (and) fatty acids lower the pH 1
- (and when) fatty acids cause the pH to be below 10 (the indicator becomes colourless) 1
- (d) observation of colour change is subjective / based on opinion  
*ignore human error unqualified*  
*ignore experimental error or examples of this* 1
- (e) bile emulsifies fats  
*allow a correct description of emulsification (i.e. breaks fat from large droplets into smaller droplets)*  
*do **not** accept a description of chemical breakdown* 1
- creates a larger surface area (of fat) 1
- (so) lipase can break down fat (to produce fatty acids) more quickly / effectively  
*allow fatty acids produced by action of lipase more quickly* 1
- [16]**
- 3.** (a) any **two** from:  
*allow proteins / hormones / antibodies / vitamins / minerals / ions / fatty acids / glycerol*
- carbon dioxide
  - water
  - glucose
  - amino acids
- ignore sugar / enzymes / nutrients / waste*
- lactic acid
- 2

(b) more haemoglobin

*max 2 marks if 'more' is not given*

1

(therefore) more oxygen can be carried / transported

1

(for) more (aerobic) respiration of muscle (cells)

**or**

more energy released for muscle (cells)

*allow less anaerobic respiration / lactic acid / oxygen*

*debt / fatigue in muscle (cells)*

*i.e. addition of 'debt'*

*do **not** accept energy produced*

1

(c) pulmonary artery

1

vena cava

1

(d) B

1

(e) any **three** from:

- arteries have a **thicker** layer of muscle (tissue) **or** veins have a **thinner** layer of muscle (tissue)\*

- arteries have a **thicker** layer of elastic tissue **or** veins have a **thinner** layer of elastic tissue\*

*\*if neither marking points 1 or 2 awarded, allow arteries have a thick wall **and** veins have a thin wall*

**or**

*arteries have a **thicker** wall or veins have a **thinner** wall for 1 mark*

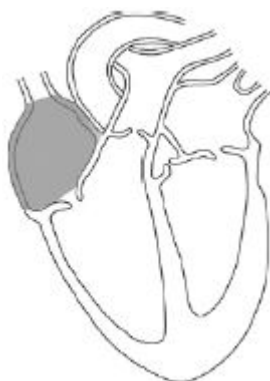
*do **not** accept 'cell wall'*

- arteries have a **narrower** lumen **or** veins have a **wider** lumen  
*allow descriptions of 'lumen'*

- arteries do not have valves **and** veins have valves  
*allow **only** veins have valves*

3

- (f) allow an X drawn anywhere in grey shaded area below:



*if a large X is drawn, award the mark if the intersection touches the grey area*

*if a label line is used, award marks if the end of the label line touches the grey area*

*allow label 'pacemaker'*

*ignore label 'right atrium'*

1

- (g) an irregular heart beat

*allow arrhythmia*

*allow fibrillation*

*ignore heart failure*

*do **not** accept cardiovascular disease / heart murmur*

1

[13]

4.

- (a) (by the guard cells) opening **and** closing the stomata

*ignore ref to guard cells being plasmolysed / turgid*

1

- (b) (water is) transported in xylem

*ignore mechanism of water entering the roots*

*do **not** accept translocation*

1

water evaporates (from leaves)

*allow loss of water vapour*

1

through the stomata

*allow between the guard cells*

*if no other marks awarded allow **1** mark for reference to transpiration*

1

(c) any **one** from:

*allow converse for plant B*

- plant **A** has more stomata  
*allow (the plants) have different numbers of stomata*
- plant **A** has more leaves  
*allow (the plants) have different numbers of leaves*
- plant **A** has bigger leaves  
*allow (the plants) have different sized leaves*
- plant **A** has a greater total surface area of leaves  
*allow (the plants) have different total surface area of leaves*  
*allow plant A has less (waxy) cuticle*  
**or**  
*(the plants) have different amounts of (waxy) cuticle*  
*allow plant A has fewer hairs on leaves*  
**or**  
*(the plants) have different number of hairs on the leaves*

1

(d)

*an answer of 10 scores 3 marks*

5.2

*allow in range 4.8 to 5.6*

1

$(5.2 \times 2 =) 10.4$

**or**

$\left(\frac{5.2}{0.5} =\right) 10.4$

*allow their calculated value in the range 8.8 to 12.0*

1

10 (cm<sup>3</sup>/hour)

*allow their calculated value in the range 8.8 to 12.0*  
*correct to 2 significant figures*

1

(e) (rate increased because)

any **two** from:

*answers must be comparative*

- (it was) warmer
- light intensity was higher
- (it was) less humid  
*allow greater water vapour gradient between leaves and environment*
- (it was) windier

2

**[10]**

5.

(a) guard (cell)

*ignore stoma / stomata*

1

(b) Species A:

- stomata open in dark / at night **or** close in light / in day

1

- stomata closed during warm(est) period **or** open when cool(er)

1

- heat (energy) / warmth increases evaporation / transpiration  
*must give explicit link between heat and transpiration*

1

- reduces water loss / evaporation / transpiration  
*ignore photosynthesis*  
*allow converse points for species B*

1

**[5]**

## Examiner reports

- 1.** Full marks were rarely seen in (b) with many students confusing or interchanging key terminology. There was frequent repetition of the words 'lock and key' without explanation; reference to 'fit' as though the word equated to 'join'; and reference to 'specific' as though the word equated to 'complementary'. In this unusual series, as some students may have begun their AS level studies, reference to activation energy was seen, although this would not usually be expected at GCSE level.

As in previous years, many students missed heating the Benedict's reagent (or only heated it to low temperatures such as 30 °C) in (d). Students' understanding of the structure of starch was generally weak, with many thinking that glucose is broken down to form starch. In addition, many students stated that photosynthesis 'creates' energy or that the glucose provides the energy for photosynthesis. Many students thought that light was needed for starch to be made. Students should be reminded to use descriptive terminology when describing variables; 'amount' is not creditworthy at this level.

- 2.**
- (a) The most common reason why students who took the right route but did not achieve full marks for this question was incorrect rounding part way through the calculation, e.g., a value of 266.6 being used rather than 266.7. It was common for students to write down a long string of decimal places in their calculations.
  - (b) Generally this question was well answered with 48% achieving three or more marks. Most students made good attempts to gain full marks and often wrote at length. The test for protein was well known by the majority of students. However, the need to describe both the starch test and the test for glucose was not recognised by a significant number of students, who often gave only the former.

In the starch test, students needed to refer to iodine solution, rather than just iodine. When describing the test for sugars, the need for heat was often omitted, with some students only referring to a water bath, rather than a hot water bath. Some students confused the colour changes for positive results.

- (c) Those students who started their answer with the idea that the lipase was breaking down fat into fatty acids (and glycerol) often went on to achieve full marks. However, many students attempted to describe their answers in terms of the neutralising effect of bile and thus resulted in no marks.
- (d) Many students gave the correct reason, referring to the subjective nature of the values. However, a significant number suggested that the measurement of time might be inaccurate due to a human error or a stop clock that was not accurate.
- (e) There were a number of students who confused the answers with that for question 07.3. It was not uncommon for students to suggest that bile is an enzyme or described their answers in terms of the pH of bile or its effects on pH of the mixture.

Those students who recognised that the difference was due to emulsification of the fat in milk often went on to describe the consequence on surface area. However, the final mark was often poorly explained or lacked detail, with students often suggesting that the bile, rather than the lipase, broke down the fat more quickly.



3.

- (a) 72% of students achieved at least one of these marks and many gained both, commonly giving two out of carbon dioxide, water and glucose. However, many students did not distinguish between substances, as required by the question, and various blood cell types were given, which were not acceptable.
- (b) There were two common omissions in answers. Firstly, many students omitted any reference to haemoglobin. Secondly many students did not link the increased oxygen supply to muscle cells, which would be needed by the athlete; and instead just indicated increased oxygen in the body, generally. A significant minority of students referred to energy being made, created or produced, which will always disqualify otherwise correct answers.
- (c) The most common error was to identify the pulmonary vein, rather than the pulmonary artery as carrying deoxygenated blood. Most students achieved at least one mark by identifying vena cava.
- (d) The correct answer 'B' was most commonly given, but the other two alternatives were occasionally seen.
- (e) This question asked for a comparison between an artery and a vein. Many students wrote about arteries and veins but did not make clear comparisons between them. Most marks were awarded for the alternate answer describing the walls of arteries as being thick(er) and the thin(ner) walls of veins. Very few students referred to the relative thicknesses of the elastic or muscle layers of the vessels.

Students who did not achieve marks often correctly stated that veins contain valves but made no further comparative statement about the lack of them in arteries. The difference in the size of the respective lumens was expressed in numerous ways; however thick or thin were terms not accepted as they don't describe the relative size of the lumens themselves.

This question would lend itself to a tabulated response thus ensuring a comparison for all answers was given but very few students took this route. Some students lost marks by referring to the cell walls of blood vessels which is clearly incorrect.

- (f) Most students added an X to the figure, but relatively few students (34%) achieved the mark. It appeared that despite efforts to clearly indicate where questions are being asked or instructions given, some students only respond when there are lines to write answers on.
- (g) There were many references to indeterminate cardiovascular diseases, with relatively few students (34%) knowing what a pacemaker is used to treat.

4.

- (a) 37% of students answered this question correctly. Many students did get confused about the difference between the guard cells and stomata, often referring to them as one and the same. References to guard cells being **in** stomata were common, as were references to the guard cells themselves closing or opening.

As the question referred to how water loss is controlled, students who referenced only the closing of the guard cells did not achieve the mark.

- (b) Generally, this question was answered well by most students and differentiated between students well. However, many students gave unnecessary information at the start of their answer, writing in detail about how water enters the roots from the soil and about the process of osmosis. Many confused the processes of osmosis and active transport here, despite this information not being relevant to the question.

Most students achieved marking points 1 and 2, however, some students confused xylem with phloem and transpiration with translocation. A number of students gave very detailed descriptions of transpiration pull and the structure of the xylem vessels.

- (c) 28% of students achieved the mark for this question. Many students who did not achieve the mark appeared to fail to read the information carefully enough, giving responses about differing conditions surrounding each plant despite the fact that the question told students the plants were kept in the same room. Often, vague responses referring to plants being better adapted were given, but with no reference as to how. Those students, who stated that plant A was bigger, failed to achieve the mark as no reference to leaf size or number was given.
- (d) The first two mark points of this calculation question were accessed well by most students. A majority of students forgot to convert their answer to 2 significant figures; students are encouraged to re-read calculation questions once they have completed their answer to ensure that all instructions have been followed.

Some students failed to read accurately from the graph (to within half a small square) and therefore were often limited to one mark (or zero if their readings were very inaccurate).

- (e) This question was generally well answered by students. A minority failed to give comparative answers, and some were confused by the concept of humidity.

5.

- (a) The majority of candidates recognised that cell **X** was a guard cell. Common incorrect answers included stoma, chloroplast and cytoplasm.
- (b) Most candidates appreciated from the data that plant species **A** opened its stomata in the dark and closed them during the light period of the day. Candidates also knew that this would help it to reduce water loss in the hot, dry desert, especially as light could be associated with a higher temperature during the daytime. Relatively few, about a fifth, stressed, unambiguously, the relationship between heat and increased evaporation of water. Some candidates, unable to think of more than one or two points, repeated them several times, in slightly different words, in order to fill the space available. A few candidates misinterpreted the data and wrote about the number of stomata open rather than about how wide the opening was.